PMAI-EG

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Predictive Maintenance Edge AI Gateway

Industrial Edge Analytics / Condition Monitoring | Rev A1 - Documentation Baseline 12 Jun 2026 | Architecture, Atomic Requirements, Interfaces & Verification

Mission profile. The gateway performs real-time machine-learning inference on high-frequency vibration data for predictive maintenance in heavy manufacturing and CNC environments. It acquires vibration deterministically, preserves data-quality context, executes approved TensorFlow Lite or ONNX models on an NPU, and sends maintenance alerts through redundant channels. Machine protection and actuation remain external unless deliberately allocated by a separate safety architecture.

Engineering baseline. The platform uses a multi-core SoC with dedicated NPU acceleration, 24 V DC industrial power and a fanless DIN-rail IP20 aluminum enclosure. OPC UA and MQTT are protected through certificate-based authentication, while secure boot/update, model identity, preprocessing compatibility and device credentials form the product trust boundary. The 15-year baseline supports OTA deployment of refined models as fleet data improves, but every new model remains a controlled release artifact requiring correlation and applicability evidence.

Definition	Controlled product interpretation
Product class	Fanless industrial edge gateway for real-time machine-learning inference on high-frequency vibration data in heavy manufacturing and CNC applications.
Safety / assurance	Watchdog-supervised monitoring and redundant alert channels; analytics results are advisory unless a separately approved machine-safety concept allocates control authority.
Documentation baseline	12 Jun 2026 - controlled PRD baseline date requested for this product.
Primary architecture	Multi-core SoC with dedicated NPU, deterministic vibration acquisition, TensorFlow Lite / ONNX model runtime, 24 V DC input, OPC UA / MQTT connectivity and certificate-based authentication.
Release philosophy	Sensor fidelity, model provenance, inference correlation, cybersecurity, manufacturing and lifecycle evidence are configuration controlled; missing evidence blocks the affected release.

Future sensor variants, model architectures, NPU/SoC changes, network stacks, plant integrations and cloud services are admitted only through controlled compatibility criteria, explicit authority boundaries and targeted regression of the affected acquisition, inference, security and alert mechanisms.

PRODUCT GOVERNANCE, MODEL AUTHORITY & REQUIREMENT QUALITY

The gateway separates deterministic vibration acquisition, machine-learning inference, maintenance alerting and plant-network services into explicit ownership domains. Model outputs support condition monitoring and maintenance decisions; they do not silently become machine-protection commands unless a separately approved plant safety concept allocates that authority.

A high-confidence model result can still be wrong because of sensor faults, distribution shift, preprocessing changes or stale calibration. Governance therefore binds every released model to sensor configuration, preprocessing version, feature definition, confidence policy and approved deployment scope.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-001	The released configuration shall enforce the following functional allocation: The gateway separates deterministic vibration acquisition, machine-learning inference, maintenance alerting and plant-network services into explicit ownership domains.	Controlled requirement set, model manifest, parameter/TBC register, interface data contract and release-evidence matrix.
PMAIEG-002	A high-confidence model result can still be wrong because of sensor faults, distribution shift, preprocessing changes or stale calibration. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Controlled requirement set, model manifest, parameter/TBC register, interface data contract and release-evidence matrix.
PMAIEG-003	Before accepting a modification to model family, sensor type, feature pipeline, alert policy, deployment scope or plant integration, the program shall close the affected analysis and Controlled requirement set, model manifest, parameter/TBC register, interface data contract and release-evidence matrix.	Controlled requirement set, model manifest, parameter/TBC register, interface data contract and release-evidence matrix.

Verification, margin and future-change rule

Changes affecting model family, sensor type, feature pipeline, alert policy, deployment scope or plant integration require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

DEPLOYMENT BOUNDARY, MACHINE INTERFACES & EXPLICIT EXCLUSIONS

The product boundary includes vibration acquisition, local compute/NPU, persistent storage, 24 V DC input, cabinet network interfaces and certificate-protected connectivity. Machine shutdown, PLC safety functions and maintenance work orders remain external unless explicitly allocated by the integrator.

Unclear ownership can cause analytics to be mistaken for a safety controller or create an unintended actuation path. The design therefore identifies every signal as measurement, metadata, diagnostic, configuration or outbound alert and prevents unapproved command authority.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-004	The product boundary includes vibration acquisition, local compute/NPU, persistent storage, 24 V DC input, cabinet network interfaces and certificate-protected connectivity. This behavior shall be preserved for every released variant to which this section applies.	System boundary diagram, ICD, installation responsibility matrix and negative-path interface review.
PMAIEG-005	When the following failure condition occurs - Unclear ownership can cause analytics to be mistaken for a safety controller or create an unintended actuation path - the system shall preserve diagnosability and prevent unqualified normal operation.	System boundary diagram, ICD, installation responsibility matrix and negative-path interface review.
PMAIEG-006	The verification record for this mechanism shall contain System boundary diagram, ICD, installation responsibility matrix and negative-path interface review.	System boundary diagram, ICD, installation responsibility matrix and negative-path interface review.

Verification, margin and future-change rule

Reopen this verification topic whenever machine interface, PLC connection, network topology, cabinet power or service interface changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

MISSION PROFILE, OPERATING MODES & SERVICE STATES

Operating modes cover power-off, boot, sensor acquisition, model warm-up, normal inference, network-disconnected buffering, degraded inference, software/model update, commissioning and authorized service. State ownership remains local even when cloud connectivity is unavailable.

A partially initialized model, stale sensor baseline or interrupted update can create plausible but invalid health scores. The state machine therefore gates inference and alert publication on declared readiness and data-validity conditions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-007	The product shall satisfy this controlled functional requirement: Operating modes cover power-off, boot, sensor acquisition, model warm-up, normal inference, network-disconnected buffering, degraded inference, software/model update, commissioning and authorized service.	State-transition tests, restart tests, disconnected-network scenarios and mode-dependent output review.
PMAIEG-008	The fault model for this topic includes: A partially initialized model, stale sensor baseline or interrupted update can create plausible but invalid health scores. Continued normal authority requires that the relevant validity checks remain satisfied.	State-transition tests, restart tests, disconnected-network scenarios and mode-dependent output review.
PMAIEG-009	Changes involving boot sequence, update flow, service-state permissions or alert publication rule shall be assessed against State-transition tests, restart tests, disconnected-network scenarios and mode-dependent output review before the configuration is accepted.	State-transition tests, restart tests, disconnected-network scenarios and mode-dependent output review.

Verification, margin and future-change rule

The change-control trigger here is any modification to boot sequence, update flow, service-state permissions or alert publication rule. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

SYSTEM ARCHITECTURE, SoC / NPU PARTITIONING & DATA FLOW

A multi-core SoC hosts deterministic acquisition and supervisory functions while a dedicated NPU executes approved inference workloads. Data flow is partitioned from sensor input through preprocessing, feature/model input, inference result, persistence and network publication with bounded queues at each handoff.

Resource starvation in the NPU or application containers must not corrupt sampling chronology or erase evidence of dropped data. Compute, memory and DMA ownership are therefore reviewed as system resources rather than assumed software details.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-010	For all applicable operating states, the mandatory product behavior is: A multi-core SoC hosts deterministic acquisition and supervisory functions while a dedicated NPU executes approved inference workloads.	Architecture review, resource-budget analysis, stress test and queue-overflow fault injection.
PMAIEG-011	Normal operation shall not be maintained without qualification when this mechanism is present: Resource starvation in the NPU or application containers must not corrupt sampling chronology or erase evidence of dropped data.	Architecture review, resource-budget analysis, stress test and queue-overflow fault injection.
PMAIEG-012	The applicable margin and behavior shall be demonstrated through Architecture review, resource-budget analysis, stress test and queue-overflow fault injection.	Architecture review, resource-budget analysis, stress test and queue-overflow fault injection.

Verification, margin and future-change rule

For future variants, a change in SoC, NPU, memory map, DMA topology, container runtime or accelerator driver is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

24 V DC INPUT, PROTECTION & INDUSTRIAL POWER INTERFACE

The released power front end accepts the program-defined 24 V industrial supply envelope and protects the gateway against polarity error, surge, brownout and wiring disturbances appropriate to its cabinet installation. Exact transient classes remain controlled by the deployment program.

Power disturbances can reset compute while leaving storage or sensors in an indeterminate state. Reset thresholds, hold-up behavior and restart sequencing are coordinated with data integrity and watchdog supervision.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-013	The implemented function shall preserve this behavior: The released power front end accepts the program-defined 24 V industrial supply envelope and protects the gateway against polarity error, surge, brownout and wiring disturbances appropriate to its cabinet installation.	Input-transient test, reverse-polarity test, brownout sweep and restart-data-integrity evidence.
PMAIEG-014	The controller shall recognize or bound the following invalid condition: Power disturbances can reset compute while leaving storage or sensors in an indeterminate state. Response priority shall follow the mechanism-specific safety or data-validity rule.	Input-transient test, reverse-polarity test, brownout sweep and restart-data-integrity evidence.
PMAIEG-015	Targeted regression for this topic shall use Input-transient test, reverse-polarity test, brownout sweep and restart-data-integrity evidence whenever input protector, DC/DC stage, connector, fuse concept or cabinet transient requirement changes.	Input-transient test, reverse-polarity test, brownout sweep and restart-data-integrity evidence.

Verification, margin and future-change rule

Changes to input protector, DC/DC stage, connector, fuse concept or cabinet transient requirement are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

POWER SEQUENCING, BROWNOUT & DATA-INTEGRITY BEHAVIOR

Power-good, reset and rail sequencing preserve valid acquisition and storage state across startup and controlled shutdown. The system records interrupted windows so missing vibration data cannot be misinterpreted as a healthy machine condition.

Marginal supply voltage can create repeated boot loops or incomplete eMMC writes. The product therefore treats undervoltage as a power-integrity condition and suspends model publication before storage integrity is jeopardized.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-016	This function is part of the released product baseline: Power-good, reset and rail sequencing preserve valid acquisition and storage state across startup and controlled shutdown.	Programmable supply sweep, repeated power-cycle testing and filesystem/database consistency inspection.
PMAIEG-017	This condition is explicitly outside valid normal behavior: Marginal supply voltage can create repeated boot loops or incomplete eMMC writes. The product shall make the loss of validity observable.	Programmable supply sweep, repeated power-cycle testing and filesystem/database consistency inspection.
PMAIEG-018	The engineering evidence set shall include Programmable supply sweep, repeated power-cycle testing and filesystem/database consistency inspection and identify the exact released variant.	Programmable supply sweep, repeated power-cycle testing and filesystem/database consistency inspection.

Verification, margin and future-change rule

If PMIC, reset threshold, storage technology, shutdown sequence or hold-up capacitance changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

VIBRATION SENSOR INTERFACE, EXCITATION & SIGNAL INTEGRITY

Sensor interfaces support the released accelerometer or vibration-transducer variants with controlled excitation, biasing, grounding, shielding and input-range assumptions. Sensor identity and orientation are part of the model applicability record.

Open sensor, short circuit, saturation, mounting looseness and cable noise can mimic bearing or tool anomalies. The acquisition chain therefore provides electrical plausibility and saturation diagnostics before data is admitted to inference.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-019	The product-level allocation for this topic is: Sensor interfaces support the released accelerometer or vibration-transducer variants with controlled excitation, biasing, grounding, shielding and input-range assumptions.	Sensor simulator, open/short tests, amplitude sweep, mounting-correlation test and noise-floor characterization.
PMAIEG-020	The design shall prevent silent acceptance of this failure mechanism: Open sensor, short circuit, saturation, mounting looseness and cable noise can mimic bearing or tool anomalies.	Sensor simulator, open/short tests, amplitude sweep, mounting-correlation test and noise-floor characterization.
PMAIEG-021	Revalidation after changes to sensor technology, excitation, cable, mounting method, orientation or input protection shall close Sensor simulator, open/short tests, amplitude sweep, mounting-correlation test and noise-floor characterization.	Sensor simulator, open/short tests, amplitude sweep, mounting-correlation test and noise-floor characterization.

Verification, margin and future-change rule

Variant evolution involving sensor technology, excitation, cable, mounting method, orientation or input protection requires a traceable change-impact review that identifies what remains valid and what must be reverified.

ANALOG FRONT END, ANTI-ALIAS FILTERING & ADC RANGE

The analog front end and ADC preserve the vibration bandwidth required by the released model while limiting alias energy and maintaining headroom for credible transients. Gain, filter corner, ADC reference and clipping behavior are configuration-controlled.

A changed filter or gain can shift model features without any software defect. Electrical transfer-function changes therefore reopen model correlation rather than being accepted only through schematic review.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-022	The controller shall implement the following behavior without transferring authority outside the declared boundary: The analog front end and ADC preserve the vibration bandwidth required by the released model while limiting alias energy and maintaining headroom for credible transients.	Frequency-response measurement, ADC linearity/range test, clipping test and model-feature regression.
PMAIEG-023	If this mechanism occurs - A changed filter or gain can shift model features without any software defect - the product shall retain a deterministic and diagnosable response.	Frequency-response measurement, ADC linearity/range test, clipping test and model-feature regression.
PMAIEG-024	The function shall not be considered verified without Frequency-response measurement, ADC linearity/range test, clipping test and model-feature regression.	Frequency-response measurement, ADC linearity/range test, clipping test and model-feature regression.

Verification, margin and future-change rule

The verification baseline is reopened when AFE gain, filter network, ADC, reference, sensor bandwidth or sampling rate changes enough to alter the stated behavior, margin, interface or failure response.

DETERMINISTIC SAMPLING, DMA & DATA-LOSS ACCOUNTING

Sensor sampling runs independently of NPU inference timing. Hardware timers and DMA or equivalent deterministic mechanisms maintain sample chronology, and every overflow, missed block or queue drop is counted and exposed with the affected time interval.

Silent sample loss can create misleading spectral features. Inference is suppressed or marked invalid whenever loss exceeds the approved data-completeness rule for that model.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-025	The released architecture shall maintain this function across its applicable modes: Sensor sampling runs independently of NPU inference timing.	Worst-case CPU/NPU load test, DMA saturation test, injected queue overflow and timestamp continuity analysis.
PMAIEG-026	The released fault handling shall cover the following case: Silent sample loss can create misleading spectral features. Normal authority shall not persist by assumption alone.	Worst-case CPU/NPU load test, DMA saturation test, injected queue overflow and timestamp continuity analysis.
PMAIEG-027	Evidence reuse after a change to sample rate, DMA channel, buffer depth, interrupt scheme or processor load is permitted only where Worst-case CPU/NPU load test, DMA saturation test, injected queue overflow and timestamp continuity analysis remains representative.	Worst-case CPU/NPU load test, DMA saturation test, injected queue overflow and timestamp continuity analysis.

Verification, margin and future-change rule

Any modification of sample rate, DMA channel, buffer depth, interrupt scheme or processor load is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

TIMEBASE, TIMESTAMPING & MULTI-SENSOR SYNCHRONIZATION

Acquisition timestamps identify the time domain used for raw data, features, inference results and alerts. Multi-channel or multi-sensor variants define synchronization accuracy and drift bounds as controlled parameters rather than assuming network time is always available.

Time jumps or unsynchronized channels can corrupt order-tracking and cross-axis features. The gateway detects discontinuity and marks affected data windows before inference.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-028	The operational requirement for this mechanism is: Acquisition timestamps identify the time domain used for raw data, features, inference results and alerts.	Time-step injection, drift test, multi-channel phase correlation and restart-timebase verification.
PMAIEG-029	The function shall fail in a controlled manner for this scenario: Time jumps or unsynchronized channels can corrupt order-tracking and cross-axis features.	Time-step injection, drift test, multi-channel phase correlation and restart-timebase verification.
PMAIEG-030	Verification of this function shall include: Time-step injection, drift test, multi-channel phase correlation and restart-timebase verification.	Time-step injection, drift test, multi-channel phase correlation and restart-timebase verification.

Verification, margin and future-change rule

Future changes in clock source, synchronization method, timestamp API or multi-sensor topology are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

PREPROCESSING PIPELINE, SCALING & WINDOWING

Preprocessing applies the released detrending, normalization, resampling, windowing and scaling operations in a versioned pipeline tied to the deployed model. Numeric precision and edge handling are included in the model interface contract.

A preprocessing library update can alter inference without changing the model file. Pipeline code and parameters therefore share the same release genealogy as model weights.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-031	The product shall retain the following controlled behavior after startup and during normal operation: Preprocessing applies the released detrending, normalization, resampling, windowing and scaling operations in a versioned pipeline tied to the deployed model.	Golden-vector comparison, fixed-point/float correlation, edge-window tests and pipeline-version traceability.
PMAIEG-032	The design shall explicitly handle this failure mechanism: A preprocessing library update can alter inference without changing the model file. The condition shall be diagnosed or bounded before normal authority is retained.	Golden-vector comparison, fixed-point/float correlation, edge-window tests and pipeline-version traceability.
PMAIEG-033	A change to preprocessing algorithm, numeric precision, library, scaling constants or window definition shall trigger impact analysis and targeted regression using Golden-vector comparison, fixed-point/float correlation, edge-window tests and pipeline-version traceability.	Golden-vector comparison, fixed-point/float correlation, edge-window tests and pipeline-version traceability.

Verification, margin and future-change rule

Engineering change control for preprocessing algorithm, numeric precision, library, scaling constants or window definition preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

FEATURE EXTRACTION & PHYSICS-BASED SIGNAL METRICS

Where the model consumes engineered features, the gateway computes controlled spectral, envelope, RMS, crest-factor, kurtosis, order or other approved metrics from valid acquisition windows. Feature units and calibration dependencies are explicit.

Feature drift caused by calibration, speed variation or changed FFT settings can masquerade as equipment degradation. The system retains enough metadata to reproduce each published feature set.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-034	The approved implementation shall meet this function-level statement: Where the model consumes engineered features, the gateway computes controlled spectral, envelope, RMS, crest-factor, kurtosis, order or other approved metrics from valid acquisition windows.	Known-signal feature test, calibration sensitivity analysis and offline reproduction of recorded feature vectors.
PMAIEG-035	The product shall not accept the following condition as normal valid operation: Feature drift caused by calibration, speed variation or changed FFT settings can masquerade as equipment degradation. It shall enter the section-defined invalid, restricted or protective response.	Known-signal feature test, calibration sensitivity analysis and offline reproduction of recorded feature vectors.
PMAIEG-036	Objective release evidence for the stated limits and assumptions shall include Known-signal feature test, calibration sensitivity analysis and offline reproduction of recorded feature vectors.	Known-signal feature test, calibration sensitivity analysis and offline reproduction of recorded feature vectors.

Verification, margin and future-change rule

A change to feature set, FFT implementation, window length, speed reference or calibration coefficient shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

NPU EXECUTION, MODEL RUNTIME & RESOURCE ISOLATION

The NPU executes only approved model artifacts through a controlled runtime. Memory allocation, accelerator context, timeout behavior and recovery are bounded so an inference stall cannot block acquisition or watchdog supervision.

Unsupported operators, runtime faults or accelerator resets can produce missing or stale outputs. Such conditions are diagnosed explicitly and do not reuse the previous inference as if it were current.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-037	The released configuration shall enforce the following functional allocation: The NPU executes only approved model artifacts through a controlled runtime.	Runtime fault injection, unsupported-operator test, NPU reset recovery and concurrent-load stress test.
PMAIEG-038	Unsupported operators, runtime faults or accelerator resets can produce missing or stale outputs. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Runtime fault injection, unsupported-operator test, NPU reset recovery and concurrent-load stress test.
PMAIEG-039	Before accepting a modification to NPU firmware, inference runtime, operator set, accelerator driver or memory allocation, the program shall close the affected analysis and Runtime fault injection, unsupported-operator test, NPU reset recovery and concurrent-load stress test.	Runtime fault injection, unsupported-operator test, NPU reset recovery and concurrent-load stress test.

Verification, margin and future-change rule

Changes affecting NPU firmware, inference runtime, operator set, accelerator driver or memory allocation require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

TENSORFLOW LITE / ONNX MODEL PACKAGING & COMPATIBILITY

TensorFlow Lite and ONNX are supported as released model packaging options only when the exact operator set, quantization, input/output tensor contract and runtime version are validated for the target hardware.

Format conversion can change numerical behavior even when model intent is unchanged. Converted artifacts therefore require correlation against the approved reference model before deployment.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-040	TensorFlow Lite and ONNX are supported as released model packaging options only when the exact operator set, quantization, input/output tensor contract and runtime version are validated for the target hardware. This behavior shall be preserved for every released variant to which this section applies.	Model-format compatibility matrix, tensor-contract test and reference-versus-target numerical correlation.
PMAIEG-041	When the following failure condition occurs - Format conversion can change numerical behavior even when model intent is unchanged - the system shall preserve diagnosability and prevent unqualified normal operation.	Model-format compatibility matrix, tensor-contract test and reference-versus-target numerical correlation.
PMAIEG-042	The verification record for this mechanism shall contain Model-format compatibility matrix, tensor-contract test and reference-versus-target numerical correlation.	Model-format compatibility matrix, tensor-contract test and reference-versus-target numerical correlation.

Verification, margin and future-change rule

Reopen this verification topic whenever model format, converter, quantization, operator implementation or runtime version changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

MODEL IDENTITY, SIGNING, VERSIONING & PROVENANCE

Every deployed model carries a unique identity, version, training/provenance reference, compatible preprocessing version, sensor applicability and cryptographic integrity record. The gateway exposes the active identity in diagnostics and telemetry.

An unknown or partially provisioned model is not treated as a benign configuration. The gateway blocks inference publication until model identity and compatibility checks pass.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-043	The product shall satisfy this controlled functional requirement: Every deployed model carries a unique identity, version, training/provenance reference, compatible preprocessing version, sensor applicability and cryptographic integrity record.	Manifest inspection, signature/integrity test, incompatible-model test and traceability audit.
PMAIEG-044	The fault model for this topic includes: An unknown or partially provisioned model is not treated as a benign configuration. Continued normal authority requires that the relevant validity checks remain satisfied.	Manifest inspection, signature/integrity test, incompatible-model test and traceability audit.
PMAIEG-045	Changes involving model weights, training lineage, manifest schema, signing key or compatibility rules shall be assessed against Manifest inspection, signature/integrity test, incompatible-model test and traceability audit before the configuration is accepted.	Manifest inspection, signature/integrity test, incompatible-model test and traceability audit.

Verification, margin and future-change rule

The change-control trigger here is any modification to model weights, training lineage, manifest schema, signing key or compatibility rules. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

CONFIDENCE, UNCERTAINTY & OUT-OF-DISTRIBUTION HANDLING

Health classification includes the released confidence or uncertainty semantics and, where provided by the approved model, out-of-distribution indicators. Thresholds are machine/fleet configuration parameters with explicit ownership.

Low-confidence or out-of-scope data must not be converted into an unjustified precise maintenance diagnosis. The gateway degrades the result to unknown, advisory or data-quality status according to the approved policy.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-046	For all applicable operating states, the mandatory product behavior is: Health classification includes the released confidence or uncertainty semantics and, where provided by the approved model, out-of-distribution indicators.	Boundary-dataset test, confidence-threshold sweep, out-of-distribution scenarios and alert-policy review.
PMAIEG-047	Normal operation shall not be maintained without qualification when this mechanism is present: Low-confidence or out-of-scope data must not be converted into an unjustified precise maintenance diagnosis.	Boundary-dataset test, confidence-threshold sweep, out-of-distribution scenarios and alert-policy review.
PMAIEG-048	The applicable margin and behavior shall be demonstrated through Boundary-dataset test, confidence-threshold sweep, out-of-distribution scenarios and alert-policy review.	Boundary-dataset test, confidence-threshold sweep, out-of-distribution scenarios and alert-policy review.

Verification, margin and future-change rule

For future variants, a change in confidence method, threshold, machine population, model retraining or decision policy is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

INFERENCE UNAVAILABLE / DEGRADED-MODE FALLBACK

If the NPU, model runtime or model artifact is unavailable, deterministic acquisition and data-quality monitoring continue where power and hardware permit. The product reports loss of analytics separately from loss of sensing.

Fallback behavior avoids both false reassurance and unnecessary machine interruption. No historical prediction is silently carried forward as a fresh result.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-049	The implemented function shall preserve this behavior: If the NPU, model runtime or model artifact is unavailable, deterministic acquisition and data-quality monitoring continue where power and hardware permit.	NPU-disable test, corrupted-model test, degraded-mode telemetry review and recovery sequencing.
PMAIEG-050	The controller shall recognize or bound the following invalid condition: Fallback behavior avoids both false reassurance and unnecessary machine interruption. Response priority shall follow the mechanism-specific safety or data-validity rule.	NPU-disable test, corrupted-model test, degraded-mode telemetry review and recovery sequencing.
PMAIEG-051	Targeted regression for this topic shall use NPU-disable test, corrupted-model test, degraded-mode telemetry review and recovery sequencing whenever fallback policy, NPU availability, model-load sequence or reporting interface changes.	NPU-disable test, corrupted-model test, degraded-mode telemetry review and recovery sequencing.

Verification, margin and future-change rule

Changes to fallback policy, NPU availability, model-load sequence or reporting interface are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

ANOMALY SCORING, TRENDING & MAINTENANCE INTERPRETATION

Anomaly scores and trends preserve model identity, machine identity, operating context and timestamp so maintenance systems can distinguish persistent degradation from transient operating changes. Aggregation windows are controlled by the application profile.

Trend logic can amplify bias if machine operating state changes. Context filters and missing-data rules prevent low-load or unrepresentative periods from being merged without visibility.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-052	This function is part of the released product baseline: Anomaly scores and trends preserve model identity, machine identity, operating context and timestamp so maintenance systems can distinguish persistent degradation from transient operating changes.	Synthetic trend test, operating-state transition cases and replay of labeled maintenance datasets.
PMAIEG-053	This condition is explicitly outside valid normal behavior: Trend logic can amplify bias if machine operating state changes. The product shall make the loss of validity observable.	Synthetic trend test, operating-state transition cases and replay of labeled maintenance datasets.
PMAIEG-054	The engineering evidence set shall include Synthetic trend test, operating-state transition cases and replay of labeled maintenance datasets and identify the exact released variant.	Synthetic trend test, operating-state transition cases and replay of labeled maintenance datasets.

Verification, margin and future-change rule

If trend window, operating-state filter, fleet threshold or maintenance interpretation changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

ALERT GENERATION, PRIORITY & REDUNDANT CHANNELS

Alerts are generated only from valid inference or deterministic diagnostic conditions and carry severity, confidence/data quality, machine identity and timestamp. Redundant channels are independent enough that one communication failure does not silently erase a critical maintenance alert.

Duplicate or conflicting alerts can overload operators. De-duplication and acknowledgement semantics are defined per receiving system without suppressing a new worsening condition.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-055	The product-level allocation for this topic is: Alerts are generated only from valid inference or deterministic diagnostic conditions and carry severity, confidence/data quality, machine identity and timestamp.	Channel-failure test, duplicate-alert test, severity mapping review and end-to-end notification evidence.
PMAIEG-056	The design shall prevent silent acceptance of this failure mechanism: Duplicate or conflicting alerts can overload operators.	Channel-failure test, duplicate-alert test, severity mapping review and end-to-end notification evidence.
PMAIEG-057	Revalidation after changes to alert threshold, severity scheme, channel provider or acknowledgement behavior shall close Channel-failure test, duplicate-alert test, severity mapping review and end-to-end notification evidence.	Channel-failure test, duplicate-alert test, severity mapping review and end-to-end notification evidence.

Verification, margin and future-change rule

Variant evolution involving alert threshold, severity scheme, channel provider or acknowledgement behavior requires a traceable change-impact review that identifies what remains valid and what must be reverified.

WATCHDOGS, HEALTH SUPERVISION & FIRST-FAULT CAPTURE

Independent watchdog supervision covers acquisition, model runtime, application services and critical communication tasks with bounded recovery ownership. First-fault data captures reset reason, software/model identity and recent health state.

A watchdog reset must not erase the reason for lost monitoring. Repeated recovery attempts are rate-limited or escalated according to the service policy.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-058	The controller shall implement the following behavior without transferring authority outside the declared boundary: Independent watchdog supervision covers acquisition, model runtime, application services and critical communication tasks with bounded recovery ownership.	Task-stall injection, watchdog timing test, reset-loop scenario and retained-diagnostic verification.
PMAIEG-059	If this mechanism occurs - A watchdog reset must not erase the reason for lost monitoring - the product shall retain a deterministic and diagnosable response.	Task-stall injection, watchdog timing test, reset-loop scenario and retained-diagnostic verification.
PMAIEG-060	The function shall not be considered verified without Task-stall injection, watchdog timing test, reset-loop scenario and retained-diagnostic verification.	Task-stall injection, watchdog timing test, reset-loop scenario and retained-diagnostic verification.

Verification, margin and future-change rule

The verification baseline is reopened when watchdog source, timeout allocation, reset architecture or retained-diagnostic storage changes enough to alter the stated behavior, margin, interface or failure response.

LOCAL STORAGE, BUFFERING & POWER-LOSS CONSISTENCY

Local storage retains raw or processed vibration data, features, alerts and update artifacts according to the released retention policy. Journaling or equivalent consistency controls prevent interrupted writes from corrupting unrelated records.

Storage-full and wear-limit conditions are observable and do not silently discard the most safety- or maintenance-relevant records without policy.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-061	The released architecture shall maintain this function across its applicable modes: Local storage retains raw or processed vibration data, features, alerts and update artifacts according to the released retention policy.	Power-cut write test, full-storage test, corruption recovery and retention-policy verification.
PMAIEG-062	The released fault handling shall cover the following case: Storage-full and wear-limit conditions are observable and do not silently discard the most safety- or maintenance-relevant records without policy. Normal authority shall not persist by assumption alone.	Power-cut write test, full-storage test, corruption recovery and retention-policy verification.
PMAIEG-063	Evidence reuse after a change to storage device, filesystem, database, retention quota or write strategy is permitted only where Power-cut write test, full-storage test, corruption recovery and retention-policy verification remains representative.	Power-cut write test, full-storage test, corruption recovery and retention-policy verification.

Verification, margin and future-change rule

Any modification of storage device, filesystem, database, retention quota or write strategy is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

OPC UA INFORMATION MODEL & CLIENT/SERVER AUTHORITY

OPC UA exposure uses a controlled namespace for measurements, health state, model identity and diagnostics. Write permissions are limited to explicitly authorized configuration objects; analytics interfaces do not become an unreviewed machine-control channel.

Misconfigured roles or stale certificates can broaden authority. Access-control testing therefore covers least privilege and denied operations, not only successful connectivity.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-064	The operational requirement for this mechanism is: OPC UA exposure uses a controlled namespace for measurements, health state, model identity and diagnostics.	OPC UA namespace review, role/permission test, certificate-expiry case and negative write-attempt test.
PMAIEG-065	The function shall fail in a controlled manner for this scenario: Misconfigured roles or stale certificates can broaden authority.	OPC UA namespace review, role/permission test, certificate-expiry case and negative write-attempt test.
PMAIEG-066	Verification of this function shall include: OPC UA namespace review, role/permission test, certificate-expiry case and negative write-attempt test.	OPC UA namespace review, role/permission test, certificate-expiry case and negative write-attempt test.

Verification, margin and future-change rule

Future changes in OPC UA stack, namespace, role model, endpoint policy or certificate profile are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

MQTT TELEMETRY, TOPIC OWNERSHIP & DELIVERY SEMANTICS

MQTT publication defines topic ownership, payload schema, quality-of-service choice, retained-message policy and timestamp semantics for health data. Broker disconnects are handled through bounded buffering without inventing successful delivery.

Retained or duplicated messages can appear current after reconnection. Payload timestamps and message identity allow the backend to reject stale or replayed condition data.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-067	The product shall retain the following controlled behavior after startup and during normal operation: MQTT publication defines topic ownership, payload schema, quality-of-service choice, retained-message policy and timestamp semantics for health data.	Broker outage/reconnect test, duplicate delivery test, retained-message test and schema compatibility review.
PMAIEG-068	The design shall explicitly handle this failure mechanism: Retained or duplicated messages can appear current after reconnection. The condition shall be diagnosed or bounded before normal authority is retained.	Broker outage/reconnect test, duplicate delivery test, retained-message test and schema compatibility review.
PMAIEG-069	A change to broker, topic hierarchy, QoS, payload schema or reconnect policy shall trigger impact analysis and targeted regression using Broker outage/reconnect test, duplicate delivery test, retained-message test and schema compatibility review.	Broker outage/reconnect test, duplicate delivery test, retained-message test and schema compatibility review.

Verification, margin and future-change rule

Engineering change control for broker, topic hierarchy, QoS, payload schema or reconnect policy preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

CERTIFICATE-BASED AUTHENTICATION & DEVICE IDENTITY

OPC UA and MQTT endpoints authenticate with program-approved device/service certificates and protected private keys. Credential scope distinguishes device identity, backend service identity and commissioning authority.

Expired, revoked or mismatched credentials cause a controlled loss of network service rather than fallback to unauthenticated operation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-070	The approved implementation shall meet this function-level statement: OPC UA and MQTT endpoints authenticate with program-approved device/service certificates and protected private keys.	Mutual-authentication test, expiry/revocation cases, key-storage review and unauthorized-endpoint test.
PMAIEG-071	The product shall not accept the following condition as normal valid operation: Expired, revoked or mismatched credentials cause a controlled loss of network service rather than fallback to unauthenticated operation. It shall enter the section-defined invalid, restricted or protective response.	Mutual-authentication test, expiry/revocation cases, key-storage review and unauthorized-endpoint test.
PMAIEG-072	Objective release evidence for the stated limits and assumptions shall include Mutual-authentication test, expiry/revocation cases, key-storage review and unauthorized-endpoint test.	Mutual-authentication test, expiry/revocation cases, key-storage review and unauthorized-endpoint test.

Verification, margin and future-change rule

A change to PKI, certificate profile, trust store, key storage or backend identity shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

NETWORK SEGMENTATION, FIREWALLING & MAINTENANCE ACCESS

Plant network interfaces are segmented according to the deployment architecture, with only required services and maintenance paths enabled. Local acquisition and watchdog functions do not depend on unrestricted enterprise-network access.

An exposed service or maintenance tunnel can bypass the intended trust boundary. Port/service inventory and firewall rules are therefore part of the released configuration.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-073	The released configuration shall enforce the following functional allocation: Plant network interfaces are segmented according to the deployment architecture, with only required services and maintenance paths enabled.	Network scan, firewall rule test, segmentation review and unauthorized-service negative test.
PMAIEG-074	An exposed service or maintenance tunnel can bypass the intended trust boundary. This condition shall trigger the diagnostic or containment behavior allocated by this section instead of silent continuation.	Network scan, firewall rule test, segmentation review and unauthorized-service negative test.
PMAIEG-075	Before accepting a modification to network topology, enabled service, firewall rule, remote-access method or VLAN policy, the program shall close the affected analysis and Network scan, firewall rule test, segmentation review and unauthorized-service negative test.	Network scan, firewall rule test, segmentation review and unauthorized-service negative test.

Verification, margin and future-change rule

Changes affecting network topology, enabled service, firewall rule, remote-access method or VLAN policy require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

SECURE BOOT, SOFTWARE INTEGRITY & RECOVERY

Boot integrity protects executable software, critical configuration and model-loading components before analytics services are enabled. Recovery images are authenticated and cannot silently downgrade security policy.

Invalid firmware or corrupted boot metadata leaves the gateway in a diagnosable restricted state with no trusted analytics publication.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-076	Boot integrity protects executable software, critical configuration and model-loading components before analytics services are enabled. This behavior shall be preserved for every released variant to which this section applies.	Secure-boot positive/negative tests, recovery-image test and rollback attempt.
PMAIEG-077	When the following failure condition occurs - Invalid firmware or corrupted boot metadata leaves the gateway in a diagnosable restricted state with no trusted analytics publication - the system shall preserve diagnosability and prevent unqualified normal operation.	Secure-boot positive/negative tests, recovery-image test and rollback attempt.
PMAIEG-078	The verification record for this mechanism shall contain Secure-boot positive/negative tests, recovery-image test and rollback attempt.	Secure-boot positive/negative tests, recovery-image test and rollback attempt.

Verification, margin and future-change rule

Reopen this verification topic whenever bootloader, trust anchor, signing key, recovery image or boot policy changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

OTA SOFTWARE / MODEL UPDATE AUTHORITY & ATOMICITY

Remote updates separate software, model and configuration artifacts, verify authenticity/compatibility before activation and preserve a recoverable previous state. Update authority is independent from ordinary telemetry publication credentials where the architecture permits.

Loss of power or network during update must not produce a half-activated model/software pair. Activation occurs only after complete integrity and compatibility checks.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-079	The product shall satisfy this controlled functional requirement: Remote updates separate software, model and configuration artifacts, verify authenticity/compatibility before activation and preserve a recoverable previous state.	Interrupted-update test, wrong-target artifact test, signature failure and activation/rollback verification.
PMAIEG-080	The fault model for this topic includes: Loss of power or network during update must not produce a half-activated model/software pair. Continued normal authority requires that the relevant validity checks remain satisfied.	Interrupted-update test, wrong-target artifact test, signature failure and activation/rollback verification.
PMAIEG-081	Changes involving update service, package format, signing key, activation policy or partition layout shall be assessed against Interrupted-update test, wrong-target artifact test, signature failure and activation/rollback verification before the configuration is accepted.	Interrupted-update test, wrong-target artifact test, signature failure and activation/rollback verification.

Verification, margin and future-change rule

The change-control trigger here is any modification to update service, package format, signing key, activation policy or partition layout. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

ROLLBACK, ANTI-ROLLBACK & CONFIGURATION COMPATIBILITY

Rollback is permitted only to approved compatible artifacts and does not defeat minimum security version or data-schema constraints. Model and preprocessing rollback remain coupled where their interface contracts require it.

Automatic rollback is prevented from masking a persistent hardware or sensor fault. The event is logged with the reason and resulting active versions.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-082	For all applicable operating states, the mandatory product behavior is: Rollback is permitted only to approved compatible artifacts and does not defeat minimum security version or data-schema constraints.	Rollback matrix, minimum-version test, incompatible-pair test and audit-log verification.
PMAIEG-083	Normal operation shall not be maintained without qualification when this mechanism is present: Automatic rollback is prevented from masking a persistent hardware or sensor fault.	Rollback matrix, minimum-version test, incompatible-pair test and audit-log verification.
PMAIEG-084	The applicable margin and behavior shall be demonstrated through Rollback matrix, minimum-version test, incompatible-pair test and audit-log verification.	Rollback matrix, minimum-version test, incompatible-pair test and audit-log verification.

Verification, margin and future-change rule

For future variants, a change in minimum version, compatibility matrix, rollback trigger or version-coupling rule is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

FANLESS THERMAL PATH, NPU POWER & CABINET DERATING

The fanless chassis conducts SoC/NPU, power-conversion and storage losses to the enclosure and cabinet environment. Temperature sensors and software derating protect compute integrity without claiming an unspecified universal ambient limit.

Sustained inference load can create thermal throttling that changes latency or sampling margin. Thermal control therefore preserves acquisition first and declares reduced analytics capacity when required.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-085	The implemented function shall preserve this behavior: The fanless chassis conducts SoC/NPU, power-conversion and storage losses to the enclosure and cabinet environment.	Thermal chamber/load test, hotspot correlation, throttling observation and hot-restart verification.
PMAIEG-086	The controller shall recognize or bound the following invalid condition: Sustained inference load can create thermal throttling that changes latency or sampling margin. Response priority shall follow the mechanism-specific safety or data-validity rule.	Thermal chamber/load test, hotspot correlation, throttling observation and hot-restart verification.
PMAIEG-087	Targeted regression for this topic shall use Thermal chamber/load test, hotspot correlation, throttling observation and hot-restart verification whenever SoC/NPU power, thermal interface material, enclosure, cabinet airflow assumption or workload changes.	Thermal chamber/load test, hotspot correlation, throttling observation and hot-restart verification.

Verification, margin and future-change rule

Changes to SoC/NPU power, thermal interface material, enclosure, cabinet airflow assumption or workload are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

DIN-RAIL IP20 MECHANICAL INTEGRATION & SERVICEABILITY

The aluminum enclosure supports DIN-rail cabinet mounting with IP20 protection as defined by the baseline. Connector access, grounding, clearances and service replacement are controlled by the installation ICD.

Mounting distortion, cable strain or inadequate cabinet spacing can alter thermal or signal behavior. Mechanical qualification therefore includes the actual mounting stack and connector loads.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-088	This function is part of the released product baseline: The aluminum enclosure supports DIN-rail cabinet mounting with IP20 protection as defined by the baseline.	Mechanical fit check, connector retention test, grounding review and mounted thermal verification.
PMAIEG-089	This condition is explicitly outside valid normal behavior: Mounting distortion, cable strain or inadequate cabinet spacing can alter thermal or signal behavior. The product shall make the loss of validity observable.	Mechanical fit check, connector retention test, grounding review and mounted thermal verification.
PMAIEG-090	The engineering evidence set shall include Mechanical fit check, connector retention test, grounding review and mounted thermal verification and identify the exact released variant.	Mechanical fit check, connector retention test, grounding review and mounted thermal verification.

Verification, margin and future-change rule

If enclosure, DIN-rail clip, connector, grounding point or cabinet spacing changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

EMC, SENSOR NOISE IMMUNITY & NETWORK COEXISTENCE

EMC design protects high-gain vibration inputs, digital compute and industrial network interfaces from conducted/radiated disturbances within the program-defined environment. Shield termination and enclosure bonding are part of the signal-integrity design.

An EMC disturbance that corrupts samples but leaves software running is especially hazardous to analytics quality. Tests therefore evaluate data validity and model output, not only resets or communication errors.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-091	The product-level allocation for this topic is: EMC design protects high-gain vibration inputs, digital compute and industrial network interfaces from conducted/radiated disturbances within the program-defined environment.	Conducted/radiated immunity and emissions testing with simultaneous sensor-quality monitoring.
PMAIEG-092	The design shall prevent silent acceptance of this failure mechanism: An EMC disturbance that corrupts samples but leaves software running is especially hazardous to analytics quality.	Conducted/radiated immunity and emissions testing with simultaneous sensor-quality monitoring.
PMAIEG-093	Revalidation after changes to cable shield, filter, enclosure bond, PCB layout, network PHY or EMC class shall close Conducted/radiated immunity and emissions testing with simultaneous sensor-quality monitoring.	Conducted/radiated immunity and emissions testing with simultaneous sensor-quality monitoring.

Verification, margin and future-change rule

Variant evolution involving cable shield, filter, enclosure bond, PCB layout, network PHY or EMC class requires a traceable change-impact review that identifies what remains valid and what must be reverified.

ENVIRONMENTAL, SHOCK / VIBRATION & CABINET QUALIFICATION

Environmental qualification addresses the industrial cabinet temperature, humidity, vibration and handling profile selected by the program. Limits not supplied by the baseline remain TBC until the installation environment is approved.

Qualification evidence must reflect powered operation and post-stress sensor accuracy, storage integrity and network service rather than cosmetic inspection alone.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-094	Variant approval is conditional on the following requirement being met: Environmental qualification addresses the industrial cabinet temperature, humidity, vibration and handling profile selected by the program.	Environmental qualification matrix, powered stress test and post-stress functional/metrology retest.
PMAIEG-095	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Environmental qualification matrix, powered stress test and post-stress functional/metrology retest.
PMAIEG-096	Production release is contingent on completion of Environmental qualification matrix, powered stress test and post-stress functional/metrology retest.	Environmental qualification matrix, powered stress test and post-stress functional/metrology retest.

Verification, margin and future-change rule

The verification baseline is reopened when installation environment, enclosure, connector, PCB stack or mounting method changes enough to alter the stated behavior, margin, interface or failure response.

SENSOR CALIBRATION, CHANNEL GENEALOGY & FIELD REPLACEMENT

Calibration records identify sensor serial number, channel gain/offset, orientation, mounting context and calibration date where applicable. Replaced sensors cannot inherit the previous channel baseline without the required commissioning checks.

Calibration drift or swapped channels can create apparently valid but physically wrong features. The gateway preserves channel identity through acquisition, model input and diagnostics.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-097	The released architecture shall maintain this function across its applicable modes: Calibration records identify sensor serial number, channel gain/offset, orientation, mounting context and calibration date where applicable.	Calibration workflow audit, swapped-channel negative test and field-replacement commissioning test.
PMAIEG-098	The released fault handling shall cover the following case: Calibration drift or swapped channels can create apparently valid but physically wrong features. Normal authority shall not persist by assumption alone.	Calibration workflow audit, swapped-channel negative test and field-replacement commissioning test.
PMAIEG-099	Evidence reuse after a change to sensor, calibration method, channel map, mounting orientation or replacement procedure is permitted only where Calibration workflow audit, swapped-channel negative test and field-replacement commissioning test remains representative.	Calibration workflow audit, swapped-channel negative test and field-replacement commissioning test.

Verification, margin and future-change rule

Any modification of sensor, calibration method, channel map, mounting orientation or replacement procedure is compared with the released assumptions on this page; unproven similarity results in targeted re-analysis or retest.

MANUFACTURING EOL, BASE MODEL FLASHING & FUNCTIONAL SCREEN

End-of-line test programs the approved base software/model set and verifies power, acquisition channels, storage, network identity, NPU execution and alert path before release. Manufacturing keys and model artifacts follow controlled provisioning procedures.

Incomplete EOL evidence is a unit-release blocker; it is not represented as a runtime degraded mode. Failed units remain quarantined until disposition is approved.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-100	Engineering sign-off for this topic shall confirm: End-of-line test programs the approved base software/model set and verifies power, acquisition channels, storage, network identity, NPU execution and alert path before release.	EOL test record, programming log, model identity check, serial-number genealogy and first-pass-yield review.
PMAIEG-101	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	EOL test record, programming log, model identity check, serial-number genealogy and first-pass-yield review.
PMAIEG-102	Release approval for this topic shall require: EOL test record, programming log, model identity check, serial-number genealogy and first-pass-yield review.	EOL test record, programming log, model identity check, serial-number genealogy and first-pass-yield review.

Verification, margin and future-change rule

Future changes in EOL fixture, programming image, model package, test limit or provisioning process are accepted only after the configuration record explains evidence reuse, new risks and the regression needed for closure.

MODEL / SOFTWARE GENEALOGY & PRODUCTION CONFIGURATION CONTROL

Production genealogy links hardware revision, sensor option, SoC/NPU variant, boot/software image, model, preprocessing package, certificates and fleet/customer configuration. Field diagnostics expose enough identity to reconstruct that released combination.

Mixing a compatible-looking but unqualified model with a hardware variant is prohibited until correlation evidence is closed.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-103	The production baseline shall comply with the following release condition: Production genealogy links hardware revision, sensor option, SoC/NPU variant, boot/software image, model, preprocessing package, certificates and fleet/customer configuration.	Configuration audit, serialized build record and incompatible-combination negative test.
PMAIEG-104	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Configuration audit, serialized build record and incompatible-combination negative test.
PMAIEG-105	A change to BOM, SoC/NPU, sensor, software, model, certificate set or customer configuration shall reopen the affected release evidence; closure shall include Configuration audit, serialized build record and incompatible-combination negative test.	Configuration audit, serialized build record and incompatible-combination negative test.

Verification, margin and future-change rule

Engineering change control for BOM, SoC/NPU, sensor, software, model, certificate set or customer configuration preserves the intent of this page by linking the modified design to updated analysis, test evidence and variant identity.

DVP - ACQUISITION CHAIN, SAMPLING & SIGNAL FIDELITY

The acquisition DVP spans sensor simulator/fixture input through timestamped samples and features across frequency, amplitude, channel count, processor load and power/environmental corners selected by the program.

Coverage includes clipping, open/short sensor, sample loss, clock disturbance and recoverability; any untested released variant keeps the corresponding verification gate open.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-106	No affected variant may be released until this condition is satisfied: The acquisition DVP spans sensor simulator/fixture input through timestamped samples and features across frequency, amplitude, channel count, processor load and power/environmental corners selected by the program.	Approved acquisition DVP report with objective pass/fail data and traceability to requirements.
PMAIEG-107	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Approved acquisition DVP report with objective pass/fail data and traceability to requirements.
PMAIEG-108	Production equivalence after a change to sensor variant, AFE, sample rate, timebase, processing load or environmental envelope shall require Approved acquisition DVP report with objective pass/fail data and traceability to requirements.	Approved acquisition DVP report with objective pass/fail data and traceability to requirements.

Verification, margin and future-change rule

A change to sensor variant, AFE, sample rate, timebase, processing load or environmental envelope shall not inherit prior approval automatically; the affected evidence is reviewed for continued representativeness before release.

DVP - INFERENCE CORRELATION, PERFORMANCE & MODEL ROBUSTNESS

Inference verification compares target-device outputs with approved reference results on representative normal, degraded and known-fault datasets. Latency, confidence behavior and resource consumption are measured with the released preprocessing/model/runtime combination.

Dataset coverage and acceptance thresholds remain program-controlled; missing representative coverage blocks release rather than being hidden by generic accuracy claims.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-109	Release approval shall enforce the following condition: Inference verification compares target-device outputs with approved reference results on representative normal, degraded and known-fault datasets.	Model-correlation report, labeled dataset results, latency statistics and model-card/release record.
PMAIEG-110	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Model-correlation report, labeled dataset results, latency statistics and model-card/release record.
PMAIEG-111	The released configuration index shall reference the following objective evidence: Model-correlation report, labeled dataset results, latency statistics and model-card/release record.	Model-correlation report, labeled dataset results, latency statistics and model-card/release record.

Verification, margin and future-change rule

Changes affecting model, dataset population, preprocessing, runtime, quantization or confidence policy require documented impact analysis against the mechanism on this page; reusable evidence is retained only within its demonstrated similarity boundary.

DVP - WATCHDOG, ALERT PATH & FAULT INJECTION

Fault-injection DVP exercises sensor loss, NPU stall, storage failure, network outage, certificate failure, task deadlock and interrupted update while observing acquisition continuity, diagnostics and redundant alert behavior.

The verification objective is correct containment and diagnosability, not artificial maintenance prediction during an unavailable inference path.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-112	Fault-injection DVP exercises sensor loss, NPU stall, storage failure, network outage, certificate failure, task deadlock and interrupted update while observing acquisition continuity, diagnostics and redundant alert behavior. This condition is mandatory for the affected release.	Fault-injection campaign report, first-fault logs and alert-delivery evidence.
PMAIEG-113	Release status for this topic shall remain open until all required objective evidence is complete and approved for the exact configuration.	Fault-injection campaign report, first-fault logs and alert-delivery evidence.
PMAIEG-114	This topic remains open in the release matrix until Fault-injection campaign report, first-fault logs and alert-delivery evidence is approved.	Fault-injection campaign report, first-fault logs and alert-delivery evidence.

Verification, margin and future-change rule

Reopen this verification topic whenever watchdog allocation, alert channel, recovery policy, storage or network service changes. The responsible engineer identifies moved assumptions and closes the resulting analysis or DVP before approval.

CYBERSECURITY VERIFICATION & INDUSTRIAL SECURITY MAPPING

Cybersecurity verification covers secure boot, authenticated update, protected credentials, OPC UA/MQTT access control, network exposure and logging. IEC 62443 or other industrial-security standards are candidate program references only until scope and target security level are approved.

A security checklist alone is insufficient; exploitable paths affecting model/configuration integrity or data authenticity require closure before the affected deployment is released.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-115	The release process shall satisfy this controlled requirement: Cybersecurity verification covers secure boot, authenticated update, protected credentials, OPC UA/MQTT access control, network exposure and logging.	Threat-model review, vulnerability test, credential negative cases and program security-compliance mapping.
PMAIEG-116	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Threat-model review, vulnerability test, credential negative cases and program security-compliance mapping.
PMAIEG-117	Release evidence after modifications to security standard, target level, enabled service, trust anchor, certificate policy or update architecture shall include Threat-model review, vulnerability test, credential negative cases and program security-compliance mapping.	Threat-model review, vulnerability test, credential negative cases and program security-compliance mapping.

Verification, margin and future-change rule

The change-control trigger here is any modification to security standard, target level, enabled service, trust anchor, certificate policy or update architecture. Acceptance requires restored margin, interface consistency and objective evidence for the affected mechanism.

15-YEAR LIFECYCLE, OTA MODEL EVOLUTION & FIELD PERFORMANCE

The baseline assigns a 15-year lifecycle with OTA deployment of refined AI models as fleet data improves. Lifecycle governance tracks model effectiveness, false-alert trends, storage wear, certificate expiry, software support and hardware obsolescence without assuming unchanged machine populations.

A refined model is a new controlled artifact, not a transparent field tweak. It must re-establish applicability to sensor hardware, preprocessing and maintenance decision policy.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-118	For each released variant, the following release condition applies: The baseline assigns a 15-year lifecycle with OTA deployment of refined AI models as fleet data improves.	Lifecycle plan, field-performance dashboard, update qualification record and obsolescence register.
PMAIEG-119	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Lifecycle plan, field-performance dashboard, update qualification record and obsolescence register.
PMAIEG-120	The program shall close Lifecycle plan, field-performance dashboard, update qualification record and obsolescence register before approving the affected production variant.	Lifecycle plan, field-performance dashboard, update qualification record and obsolescence register.

Verification, margin and future-change rule

For future variants, a change in model retraining, fleet population, certificate lifetime, storage endurance or component availability is an engineering event, not an editorial update; the impact record defines evidence reuse and targeted regression.

MODEL DRIFT MONITORING, FIELD FEEDBACK & REVALIDATION TRIGGERS

Field monitoring tracks data-quality shifts, alert rates, confidence distribution and other approved model-performance indicators without assuming ground truth is continuously available. Drift metrics are diagnostic evidence used to trigger engineering review, not automatic authorization to change model thresholds in service.

A model can remain numerically stable while the machine population or operating regime changes. The lifecycle process therefore distinguishes true equipment degradation from distribution shift and documents when retraining or revalidation is required.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-121	The configuration shall not be released unless this statement is true: Field monitoring tracks data-quality shifts, alert rates, confidence distribution and other approved model-performance indicators without assuming ground truth is continuously available.	Field-performance trend review, drift-indicator analysis, labeled-event backcheck where available and revalidation trigger record.
PMAIEG-122	A missing verification result, manufacturing record or similarity justification for this topic is a release-gate condition and shall not be converted into runtime behavior.	Field-performance trend review, drift-indicator analysis, labeled-event backcheck where available and revalidation trigger record.
PMAIEG-123	Variant sign-off shall reference Field-performance trend review, drift-indicator analysis, labeled-event backcheck where available and revalidation trigger record and the applicable configuration identity.	Field-performance trend review, drift-indicator analysis, labeled-event backcheck where available and revalidation trigger record.

Verification, margin and future-change rule

Changes to machine population, operating regime, sensor mounting, process condition, model threshold or training dataset are screened against the failure mechanism described above, with only demonstrably unaffected evidence carried forward.

DIAGNOSTICS, SERVICE EXPORT & ENGINEERING DATA ACCESS

Local and remote diagnostics expose sensor quality, acquisition-loss counters, active software/model identity, inference availability, storage health, temperatures, certificate state and recent faults without granting uncontrolled modification authority. Engineering data export is explicitly permissioned and traceable.

A diagnostic path can become an unintended control or data-exfiltration channel if write authority, service credentials or export scope are not constrained. Service functions therefore follow least-privilege and do not bypass model or security release controls.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-124	This function is part of the released product baseline: Local and remote diagnostics expose sensor quality, acquisition-loss counters, active software/model identity, inference availability, storage health, temperatures, certificate state and recent faults without granting uncontrolled modification authority.	Diagnostic-service test, permission matrix, negative write attempts, export-content review and service audit logging.
PMAIEG-125	This condition is explicitly outside valid normal behavior: A diagnostic path can become an unintended control or data-exfiltration channel if write authority, service credentials or export scope are not constrained. The product shall make the loss of validity observable.	Diagnostic-service test, permission matrix, negative write attempts, export-content review and service audit logging.
PMAIEG-126	The engineering evidence set shall include Diagnostic-service test, permission matrix, negative write attempts, export-content review and service audit logging and identify the exact released variant.	Diagnostic-service test, permission matrix, negative write attempts, export-content review and service audit logging.

Verification, margin and future-change rule

If diagnostic protocol, service account, exported dataset, debug interface or maintenance tool changes, the release owner documents the affected assumptions, selects the required regression and updates the evidence index before sign-off.

SUPPLIER CHANGE, SECOND SOURCE & TECHNOLOGY REFRESH

Second-source sensors, storage, SoC/NPU devices, network PHYs or power components are evaluated for effects on signal fidelity, timing, thermal behavior, security and model correlation. Similarity claims identify exactly which evidence can be reused.

Production equivalence is not approved while affected margin analysis, model correlation or regression DVP remains incomplete.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-127	The approved baseline shall demonstrate the following before release: Second-source sensors, storage, SoC/NPU devices, network PHYs or power components are evaluated for effects on signal fidelity, timing, thermal behavior, security and model correlation.	Supplier change-impact assessment, similarity report and targeted regression package.
PMAIEG-128	Incomplete, failed or variant-mismatched evidence for this topic shall block release of the affected configuration; it shall not be represented as a fictitious runtime degraded state.	Supplier change-impact assessment, similarity report and targeted regression package.
PMAIEG-129	The release package shall contain Supplier change-impact assessment, similarity report and targeted regression package for the exact hardware/software variant.	Supplier change-impact assessment, similarity report and targeted regression package.

Verification, margin and future-change rule

Variant evolution involving critical component supplier, die revision, firmware, process or lifecycle substitution requires a traceable change-impact review that identifies what remains valid and what must be reverified.

FINAL RELEASE CLOSURE & CONFIGURATION FREEZE

Release closes only when the selected hardware/software/model/certificate configuration has complete requirements traceability, open-TBC disposition, DVP evidence, manufacturing controls, cybersecurity review and lifecycle ownership. Deviations are explicit and approved by accountable engineering authorities.

Open qualification, model-correlation, manufacturing or supplier-equivalence gaps remain release blockers; they are never converted into invented runtime behavior merely to close documentation.

Atomic product requirements

Requirement ID	Atomic requirement	Verification / release evidence
PMAIEG-130	Variant approval is conditional on the following requirement being met: Release closes only when the selected hardware/software/model/certificate configuration has complete requirements traceability, open-TBC disposition, DVP evidence, manufacturing controls, cybersecurity review and lifecycle ownership.	Final release checklist, approved deviation log, configuration index and signed evidence matrix.
PMAIEG-131	The affected hardware or software variant shall remain unreleased while required qualification or production evidence for this topic is open.	Final release checklist, approved deviation log, configuration index and signed evidence matrix.
PMAIEG-132	Production release is contingent on completion of Final release checklist, approved deviation log, configuration index and signed evidence matrix.	Final release checklist, approved deviation log, configuration index and signed evidence matrix.

Verification, margin and future-change rule

The verification baseline is reopened when any released baseline element or unresolved evidence item changes enough to alter the stated behavior, margin, interface or failure response.